

Saad Khairullah

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PROJECTS

AI Interviewer (Capstone) | github.com/vkr-vavilla/Senior_Design

Python, FastAPI, PyTorch, HuggingFace, vLLM, WebSocket, JWT, Gemini API

- Built an async **FastAPI** backend for concurrent interview sessions, with **JWT** authentication and middleware-protected API routes for secure session management, serving a **Next.js** frontend.
- Wrapped the app in a **Tauri desktop client** so users can run full interviews fully offline on consumer hardware
- Designed a **Python** data pipeline using the **Gemini API** to process 47 YouTube transcripts into **300k+ structured dialogue tokens** formatted as multi-turn conversations for fine-tuning
- Fine-tuned a 7B-parameter **Qwen** model using **LoRA/PEFT** with **PyTorch** and **HuggingFace Transformers**; deployed on a **vLLM** inference server, delivering speech-to-speech responses in under 500ms end-to-end, Ollama inference for platform compatibility

Blockheads Game Server | github.com/saadkhairullah/The-Blockheads-Server

Java, TypeScript, Python, LMDB, Linux, Ghidra, Wireshark, iptables, Unix Domain Sockets

- Extended an existing **Java proxy** by reverse engineering a closed-source **UDP/ENet protocol** with **Ghidra** and **Wireshark**; added custom protocol features including packet rerouting and application-layer malicious packet blocking, serving **300+ registered players**
- Optimized **LMDB** queries across ~6GB of player, world, and server data from **O(n)** full scans (20+ seconds) to **O(log n)** direct key lookups (under 5ms) by restructuring the storage schema
- Architected a multi-service system in **TypeScript, Java, and Python** (~40k lines) communicating via **Unix Domain Sockets**; deployed and maintain on a live **Linux** server with multi-layer **DDoS mitigation** via iptables blocking 20+ real attacks

Healthcare Voice AI Agent | github.com/saadkhairullah/Telephone-Voice-Assistant-Patient

Python, FastAPI, WebSocket, Pipecat, Telnyx, Deepgram, Silero VAD, Llama 3.3 70B

- Built solo in 31 hours: an **agentic AI system** that autonomously places outbound calls to clinic agents, simulating a patient to probe service quality across **60+ real production calls** for under \$2 in total API costs
- Orchestrated a real-time voice pipeline with **Pipecat**, integrating **Telnyx** (telephony), **Deepgram** (STT), **Silero VAD**, and **Llama 3.3 70B** to manage turn-taking and live agentic decisioning
- Built a **FastAPI WebSocket** backend handling real-time bidirectional audio streams with non-blocking async I/O

EXPERIENCE

AI Data Annotator (Freelance) – Outlier.ai (Remote)

April 2024 – July 2024

- Evaluated and ranked AI-generated **Python** and **Java** code responses on technical accuracy and quality, contributing structured feedback to **LLM** reinforcement learning pipelines

EDUCATION

University of Texas at Arlington | Arlington, TX

Graduating August 6th 2026

Bachelor of Science in Software Engineering

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Systems and File Structures, Linear Algebra, Computer Organization, Signal Processing, Object-Oriented Software Engineering

SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, C

Backend / APIs: FastAPI, Node.js, Express, Flask, Spring Boot, REST APIs, WebSocket, async I/O, JWT, middleware, IPC

Frontend: React, Next.js, MapboxGL, HTML/CSS

AI / LLM: LLM integration, agentic systems, fine-tuning, LoRA/PEFT/QLoRA, PyTorch, HuggingFace, vLLM, Ollama, Pipecat

Databases: MySQL, MongoDB, LMDB, SQL

Systems / Tools: Linux, Docker, Git, GitHub Actions, Ghidra, Wireshark, iptables, Unix Domain Sockets, Bash, Tauri
